

Monday, Apr 1

Review of Linear Combinations

Consider a regression model with parameters $\beta_0, \beta_1, \dots, \beta_k$. A *linear combination* is a *linear function* of the form

$$\ell = a_0\beta_0 + a_1\beta_1 + a_2\beta_2 + \dots + a_k\beta_k + b,$$

where $a_0, a_1, \dots, a_k$ and b are user-specified coefficients (frequently $b = 0$, and to simplify what follows assume $b = 0$). Often (but not always) quantities of interest can be expressed as linear combinations.

Example: Consider the linear model

$$E(G_i) = \beta_0 + \beta_1 d_i + \beta_2 t_i + \beta_3 d_i t_i.$$

for the `whiteside` data, where G_i is gas consumption, t_i is temperature, and d_i is an indicator variable for *after* insulation such that

$$d_i = \begin{cases} 1, & \text{if the } i\text{-th observation is after insulation,} \\ 0, & \text{otherwise.} \end{cases}$$

Then we can write the model case-wise as

$$E(G_i) = \begin{cases} \beta_0 + \beta_2 t_i, & \text{if } i\text{-th observation is before insulation,} \\ \beta_0 + \beta_1 + (\beta_2 + \beta_3) t_i, & \text{if } i\text{-th observation is after insulation.} \end{cases}$$

Several quantities that might be of interest can be written as linear combinations. The `lincon` and `contrast` functions facilitate inferences regarding linear combinations. First we specify the model.

```
m <- lm(Gas ~ Insul + Temp + Insul:Temp, data = MASS::whiteside)
summary(m)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	6.8538	0.13596	50.409	7.997e-46
InsulAfter	-2.1300	0.18009	-11.827	2.316e-16
Temp	-0.3932	0.02249	-17.487	1.976e-23
InsulAfter:Temp	0.1153	0.03211	3.591	7.307e-04

Now consider the following linear combinations.

1. Rate of change before insulation:

$$\ell = 0\beta_0 + 0\beta_1 + 1\beta_2 + 0\beta_3 = \beta_2.$$

```
library(trtools)
lincon(m, a = c(0,0,1,0))
```

	estimate	se	lower	upper	tvalue	df	pvalue
(0,0,1,0),0	-0.3932	0.02249	-0.4384	-0.3481	-17.49	52	1.976e-23

```
contrast(m,
  a = list(Insul = "Before", Temp = 1),
  b = list(Insul = "Before", Temp = 0))
```

estimate	se	lower	upper	tvalue	df	pvalue
-0.3932	0.02249	-0.4384	-0.3481	-17.49	52	1.976e-23

2. Rate of change after insulation:

$$\ell = 0\beta_0 + 0\beta_1 + 1\beta_2 + 1\beta_3 = \beta_2 + \beta_3.$$

```
lincon(m, a = c(0,0,1,1))
```

	estimate	se	lower	upper	tvalue	df	pvalue
(0,0,1,1),0	-0.2779	0.02292	-0.3239	-0.2319	-12.12	52	8.936e-17

```
contrast(m,
  a = list(Insul = "After", Temp = 1),
  b = list(Insul = "After", Temp = 0))
```

estimate	se	lower	upper	tvalue	df	pvalue
-0.2779	0.02292	-0.3239	-0.2319	-12.12	52	8.936e-17

3. Expected gas consumption before insulation at 5C:

$$\ell = 1\beta_0 + 0\beta_1 + 5\beta_2 + 0\beta_3 = \beta_0 + 5\beta_2.$$

```
lincon(m, a = c(1,0,0,5))
```

	estimate	se	lower	upper	tvalue	df	pvalue
(1,0,0,5),0	7.43	0.2671	6.894	7.966	27.82	52	6.734e-33

```
contrast(m, a = list(Insul = "Before", Temp = 5))
```

estimate	se	lower	upper	tvalue	df	pvalue
4.888	0.06383	4.76	5.016	76.57	52	3.885e-55

Technical Details

To estimate ℓ we simply replace $\beta_0, \beta_1, \dots, \beta_k$ with estimates to obtain

$$\hat{\ell} = a_0\hat{\beta}_0 + a_1\hat{\beta}_1 + \dots + a_k\hat{\beta}_k.$$

To compute the variance of $\hat{\ell}$ we use the result that

$$\text{Var}(\hat{\ell}) = \sum_{j=0}^k \sum_{j'=0}^k a_j a_{j'} \text{Cov}(\hat{\beta}_j, \hat{\beta}_{j'}),$$

where $\text{Cov}(\hat{\beta}_j, \hat{\beta}_{j'})$ is the *covariance* between the estimators $\hat{\beta}_j$ and $\hat{\beta}_{j'}$, and note that the covariance of an estimator with itself is its *variance* — i.e., $\text{Cov}(\hat{\beta}_j, \hat{\beta}_j) = \text{Var}(\hat{\beta}_j)$.

Example: If we had the model $E(Y_i) = \beta_0 + \beta_1 x_i$ then the covariances can be arranged in a *matrix* as

$$\begin{bmatrix} \text{Cov}(\hat{\beta}_0, \hat{\beta}_0) & \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ \text{Cov}(\hat{\beta}_1, \hat{\beta}_0) & \text{Cov}(\hat{\beta}_1, \hat{\beta}_1) \end{bmatrix} = \begin{bmatrix} \text{Var}(\hat{\beta}_0) & \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) \\ \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) & \text{Var}(\hat{\beta}_1) \end{bmatrix},$$

noting that the covariance of a variable with itself is its *variance*, and covariances are symmetric in the sense that $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = \text{Cov}(\hat{\beta}_1, \hat{\beta}_0)$. If we wanted to compute the variance of the linear combination $\ell = 1\beta_0 + 5\beta_1$ so that $a_0 = 1$ and $a_1 = 5$, then the variance of $\hat{\ell}$ would be

$$\text{Var}(\hat{\ell}) = 1 \times 1 \times \text{Var}(\hat{\beta}_0) + 1 \times 5 \times \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) + 5 \times 1 \times \text{Cov}(\hat{\beta}_1, \hat{\beta}_0) + 5 \times 5 \times \text{Var}(\hat{\beta}_1),$$

which simplifies to

$$\text{Var}(\hat{\ell}) = \text{Var}(\hat{\beta}_0) + 10\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) + 25\text{Var}(\hat{\beta}_1).$$

Typically we only have estimates of the variances and covariances so we can only compute an estimate of $\text{Var}(\hat{\ell})$ denoted as $\widehat{\text{Var}}(\hat{\ell})$. The square root of this estimate is the *standard error* you usually see reported by R functions like `contrast` and `lincon`. The confidence interval for ℓ is then

$$\hat{\ell} \pm t\sqrt{\widehat{\text{Var}}(\hat{\ell})} \quad \text{or} \quad \hat{\ell} \pm z\sqrt{\widehat{\text{Var}}(\hat{\ell})},$$

and the test statistic is

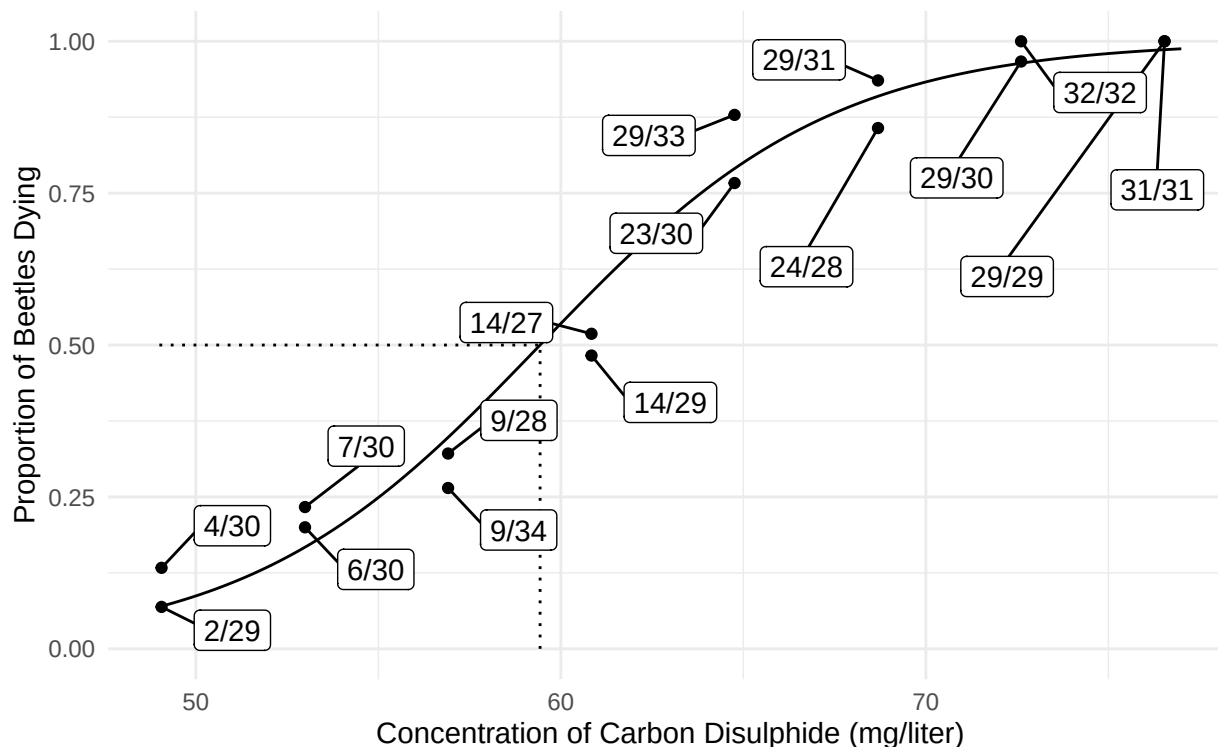
$$\frac{\hat{\ell} - \ell}{\sqrt{\widehat{\text{Var}}(\hat{\ell})}},$$

where typically $\ell = 0$ because the null hypothesis is that $H_0: \ell = 0$. This is the test statistic you typically see reported by R functions like `contrast` and `lincon`.

The Delta Method

The *delta method* is used to approximate the sampling distribution of a *nonlinear* function of model parameters.

Example: Consider a logistic regression model for the `bliss` data.



The model can be written as

$$\log \left[\frac{E(Y)}{1 - E(Y)} \right] = \beta_0 + \beta_1 x,$$

where x is concentration. The LD_{50} is the value of x (e.g., dose, concentration) such that the expected proportion of deaths is 0.5 so that

$$\log \left[\frac{0.5}{1 - 0.5} \right] = \beta_0 + \beta_1 \text{LD}_{50}.$$

Solving for LD_{50} gives

$$LD_{50} = \frac{-\beta_0}{\beta_1}.$$

An estimator of LD_{50} is

$$\widehat{LD}_{50} = \frac{-\hat{\beta}_0}{\hat{\beta}_1},$$

but this is a *nonlinear* function of $\hat{\beta}_0$ and $\hat{\beta}_1$.

Technical Details

The delta method uses the following results:

1. Nonlinear functions of the model parameters can be “locally approximated” by a *linear* function (see below).
2. As n increases sampling distributions become increasingly “concentrated” in a location near the actual quantity being estimated.
3. If the sampling distribution of $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k$ is approximately (multivariate) normal, then the sampling distribution of an (approximately) linear function of these estimators is also approximately normal.

Let $f(\beta_0, \beta_1, \dots, \beta_k)$ be a function of $\beta_0, \beta_1, \dots, \beta_k$ such as $f(\beta_0, \beta_1) = -\beta_0/\beta_1$. Using some calculus it can be shown that in many practical cases we can *approximate* $f(\beta_0, \beta_1, \dots, \beta_k)$ at any specific point (i.e., “locally”) by

$$\ell \approx c + d_0\beta_0 + d_1\beta_1 + \dots + d_k\beta_k,$$

where c is a constant. The coefficients $d_0, d_1, \dots, d_k$ are the *partial derivatives* of $f(\beta_0, \beta_1, \dots, \beta_k)$ at given values of $\beta_0, \beta_1, \dots, \beta_k$ so that

$$d_j = \frac{\partial f(\beta_0, \beta_1, \dots, \beta_k)}{\partial \beta_j}.$$

We approximate $f(\beta_0, \beta_1, \dots, \beta_k)$ at $f(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k)$. Then we can compute

$$\text{Var}[f(\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_k)] \approx \sum_{j=0}^k \sum_{j'=0}^k d_j d_{j'} \text{Cov}(\hat{\beta}_j, \hat{\beta}_{j'}),$$

where the partial derivatives are evaluated at the parameter estimates. The main mathematical challenge perhaps is evaluating the partial derivatives. This may be done *analytically* or *numerically*.

Example: Consider again the problem of making inferences regarding $LD_{50} = -\beta_0/\beta_1$. The `dmethod` function from the `trtools` package implements the delta method, using *numerically approximated* derivatives for greater flexibility.

```
library(trtools)
m <- glm(cbind(dead, exposed - dead) ~ concentration, family = binomial, data = bliss)
cbind(summary(m)$coefficients, confint(m))
```

	Estimate	Std. Error	z value	Pr(> z)	2.5 %	97.5 %
(Intercept)	-14.8084	1.28976	-11.48	1.633e-30	-17.4785	-12.4089
concentration	0.2492	0.02138	11.65	2.250e-31	0.2095	0.2935

```
dmethod(m, pfunc = "-b0/b1", pname = c("b0", "b1"))
```

estimate	se	lower	upper	tvalue	df	pvalue
59.43	0.5287	58.39	60.47	112.4	Inf	0

Example: The “instantaneous” marginal effect of dose for the logistic model for the `bliss` data is the slope of a tangent line at a given value of dose (x). A little calculus shows that this is

$$\frac{\partial}{\partial x} \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}} = \frac{\beta_1 e^{\beta_0 + \beta_1 x}}{(1 + e^{\beta_0 + \beta_1 x})^2}.$$

Here is the estimate of the instantaneous marginal effect at the (estimated) LD_{50} from above.

```
dmethod(m, pfunc = "b1*exp(b0+b1*59.43)/(1+exp(b0+b1*59.43))^2", pname = c("b0","b1"))
```

estimate	se	lower	upper	tvalue	df	pvalue
0.06229	0.005346	0.05181	0.07277	11.65	Inf	2.245e-31

The `margeff` function does the same thing but computes the derivative above numerically.

```
margeff(m, delta = 0.001,
a = list(concentration = 59.43 + 0.001),
b = list(concentration = 59.43))
```

estimate	se	lower	upper	tvalue	df	pvalue
0.06229	0.005346	0.05181	0.07277	11.65	Inf	2.247e-31

Since $LD_{50} = -\beta_0/\beta_1$ we can substitute this for x in the expression for the instantaneous marginal effect. Simplifying gives

$$\frac{\beta_1 e^{\beta_0 + \beta_1(-\beta_0/\beta_1)}}{(1 + e^{\beta_0 + \beta_1(-\beta_0/\beta_1)})^2} = \beta_1/4.$$

So then we have

```
dmethod(m, pfunc = "b1/4", pname = c("b0","b1"))
```

estimate	se	lower	upper	tvalue	df	pvalue
0.06229	0.005346	0.05181	0.07277	11.65	Inf	2.25e-31

but since $\beta_1/4 = a_0\beta_0 + a_1\beta_1$ where $a_0 = 0$ and $a_1 = 1/4$ we can do this using `lincon` as well.

```
lincon(m, a = c(0,0.25))
```

	estimate	se	lower	upper	tvalue	df	pvalue
(0,1/4),0	0.06229	0.005346	0.05181	0.07277	11.65	Inf	2.25e-31

It is useful to note that when the function is a linear combination, `dmethod` will produce the same results as `lincon`. But it provides a slightly different interface which can be useful since it does not require us to figure out the coefficients $a_0, a_1, \dots, a_k$.

Example: Now consider the *discrete* marginal effect of increasing dose from 50 to 60 mg/liter. This is

$$\frac{e^{\beta_0 + \beta_1 60}}{1 + e^{\beta_0 + \beta_1 60}} - \frac{e^{\beta_0 + \beta_1 50}}{1 + e^{\beta_0 + \beta_1 50}}.$$

This is a nonlinear function of β_0 and β_1 . We can apply the delta method as follows.

```
dmethod(m, pfunc = "plogis(b0 + b1*60) - plogis(b0 + b1*50)", pname = c("b0","b1"))
```

estimate	se	lower	upper	tvalue	df	pvalue
0.4483	0.02796	0.3935	0.5031	16.03	Inf	7.657e-58

Note that the R function `plogis` is $e^x/(1 + e^x)$ so that saves us a little typing. The `margeff` function does the same thing but is maybe easier to use here.

```
margeff(m, a = list(concentration = 60), b = list(concentration = 50))
```

estimate	se	lower	upper	tvalue	df	pvalue
0.4483	0.02796	0.3935	0.5031	16.03	Inf	7.657e-58

Example: Consider a polynomial regression model for the expected tensile strength as a function of percent hardwood such that

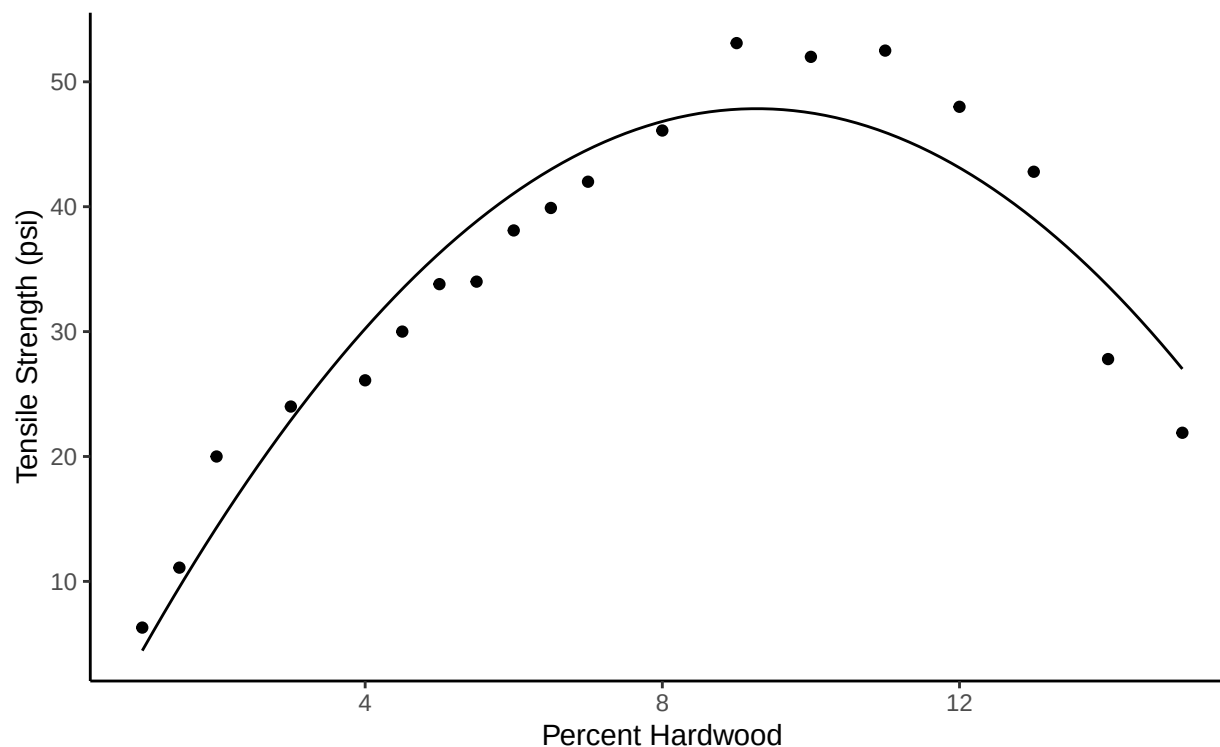
$$E(S_i) = \beta_0 + \beta_1 p_i + \beta_2 p_i^2,$$

where S_i is tensile strength and p_i is percent hardwood. Here is what the estimated model looks like.

```
library(GLMsData)
data(paper)
m <- lm(Strength ~ Hardwood + I(Hardwood^2), data = paper)
summary(m)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	-6.6742	3.39971	-1.963	6.725e-02
Hardwood	11.7640	1.00278	11.731	2.854e-09
I(Hardwood^2)	-0.6345	0.06179	-10.270	1.894e-08

```
d <- data.frame(Hardwood = seq(1, 15, length = 100))
d$yhat <- predict(m, newdata = d)
p <- ggplot(paper, aes(x = Hardwood, y = Strength)) +
  geom_point() + theme_classic() +
  geom_line(aes(y = yhat), data = d) +
  labs(x = "Percent Hardwood", y = "Tensile Strength (psi)")
plot(p)
```



Now consider the problem of estimating (a) the value of percent hardwood at which expected tensile strength is maximized, and (b) the maximum expected tensile strength. The percent hardwood that maximizes expected tensile strength (p_m) can be derived as

$$\frac{d(\beta_0 + \beta_1 p + \beta_2 p^2)}{dp} = \beta_1 + 2\beta_2 p = 0 \Rightarrow p_m = \frac{-\beta_1}{2\beta_2}.$$

Now if we replace p with p_m in $E(S) = \beta_0 + \beta_1 p + \beta_2 p^2$ we find the maximum expected tensile strength (s_m)

to be

$$s_m = \beta_0 + \beta_1 \frac{-\beta_1}{2\beta_2} + \beta_2 \left(\frac{-\beta_1}{2\beta_2} \right)^2 = \frac{4\beta_0\beta_2 - \beta_1^2}{4\beta_2}.$$

Both p_m and s_m are nonlinear functions of β_0 , β_1 and β_2 .

```
dmethod(m, pfunc = "-b1/(2*b2)", pname = c("b0", "b1", "b2"))
```

```
estimate      se lower upper tvalue  df pvalue
9.27 0.2344  8.81 9.729  39.55 Inf    0
```

```
dmethod(m, pfunc = "(4*b0*b2-b1^2)/(4*b2)", pname = c("b0", "b1", "b2"))
```

```
estimate      se lower upper tvalue  df      pvalue
47.85 1.495 44.92 50.78  32.01 Inf 7.656e-225
```

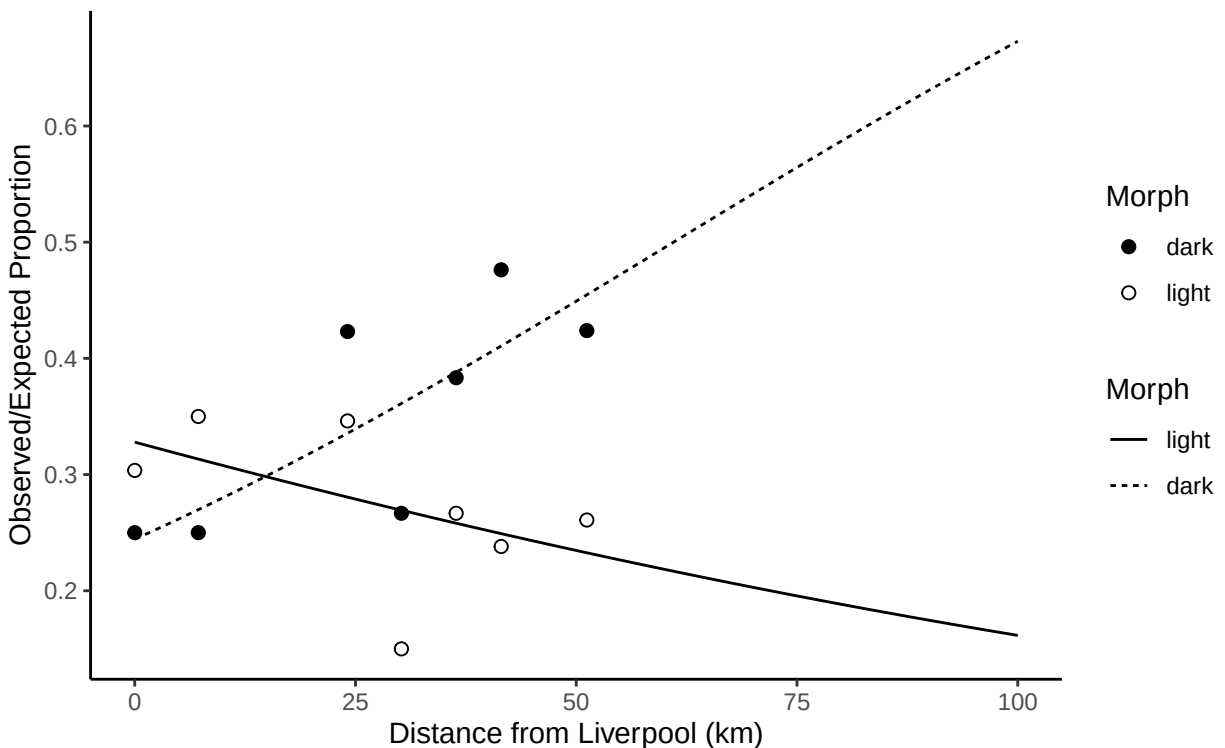
Whenever possible try to check these kinds of estimates against a plot of the data and/or model to catch errors.

Example: Consider a study on moth coloration and predation.

```
library(Sleuth3)
m <- glm(cbind(Removed, Placed - Removed) ~ Distance + Morph +
  Distance:Morph, data = case2102, family = binomial)

d <- expand.grid(Distance = seq(0, 100, length = 100), Morph = c("light", "dark"))
d$yhat <- predict(m, d, type = "response")

p <- ggplot(case2102, aes(x = Distance, y = Removed/Placed)) +
  geom_point(aes(fill = Morph), size = 2, shape = 21) +
  geom_line(aes(y = yhat, linetype = Morph), data = d) +
  scale_fill_manual(values = c("black", "white")) + theme_classic() +
  labs(y = "Observed/Expected Proportion", x = "Distance from Liverpool (km)")
plot(p)
```



Suppose we want to estimate the point at which the two curves *cross*. First consider the model parameterization.

```
summary(m)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.12899	0.197906	-5.705	1.166e-08
Distance	0.01850	0.005645	3.277	1.048e-03
Morphlight	0.41126	0.274490	1.498	1.341e-01
Distance:Morphlight	-0.02779	0.008085	-3.437	5.884e-04

This model can be written as

$$E(Y_i) = \frac{e^{\eta_i}}{1 + e^{\eta_i}},$$

where Y_i is the i -th observation of the proportion of removed moths, and the linear predictor η_i is parameterized as

$$\eta_i = \begin{cases} \beta_0 + \beta_1 d_i, & \text{if the morph of the } i\text{-th observation is dark,} \\ \beta_0 + \beta_2 + (\beta_1 + \beta_3) d_i, & \text{if the morph of the } i\text{-th observation is light,} \end{cases}$$

where d_i is the distance from Liverpool for the i -th observation. Let d_c be the distance at which the expected proportions are equal for both light and dark moths, then d_c must satisfy the equation

$$\beta_0 + \beta_1 d_c = \beta_0 + \beta_2 + (\beta_1 + \beta_3) d_c.$$

Solving this equation for d_c gives $d_c = -\beta_2/\beta_3$ (assuming that $\beta_3 \neq 0$). Note that d_c is a nonlinear function of β_2 and β_3 . Inferences concerning d_c can be made using the delta method.

```
dmethord(m, pfunc = "-b2/b3", pname = c("b0", "b1", "b2", "b3"))
```

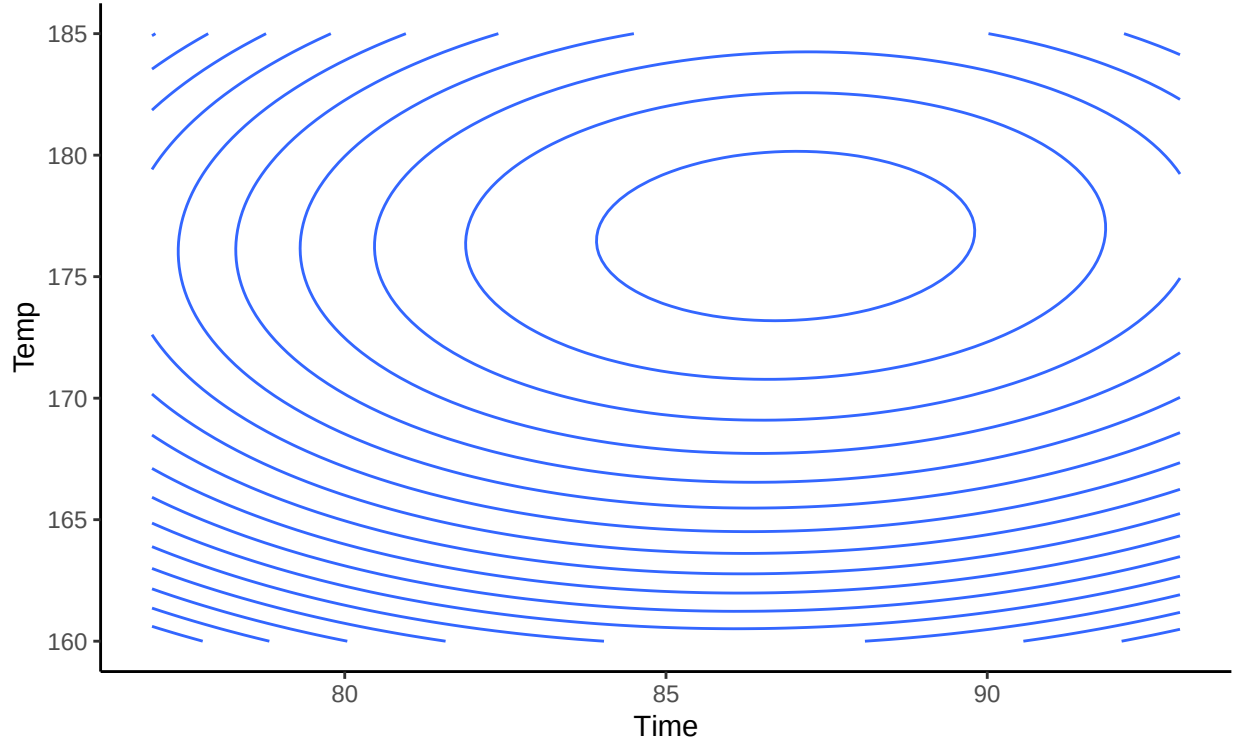
estimate	se	lower	upper	tvalue	df	pvalue
14.8	6.563	1.937	27.66	2.255	Inf	0.02413

Example: Suppose we want to estimate the values of *two* explanatory variables that maximize (or minimize) and expected response.

```
m <- lm(Yield ~ Time + Temp + I(Time^2) + I(Temp^2) +
  I(Time*Temp), data = rsm::ChemReact)

d <- expand.grid(Time = seq(77, 93, length = 100),
  Temp = seq(160, 185, length = 100))
d$yhat <- predict(m, newdata = d)

p <- ggplot(d, aes(x = Time, y = Temp, z = yhat))
p <- p + geom_contour(bins = 20) + theme_classic()
plot(p)
```

The model is

$$E(Y_i) = \beta_0 + \beta_1 \text{time}_i + \beta_2 \text{temp}_i + \beta_3 \text{time}_i^2 + \beta_4 \text{temp}_i^2 + \beta_5 \text{time}_i \text{temp}_i,$$

where Y_i is yield. Let a_m and b_m be the values of time and temperature, respectively, that maximize $E(Y)$. These satisfy

$$\begin{aligned} \frac{\partial(\beta_0 + \beta_1 a_m + \beta_2 b_m + \beta_3 a_m^2 + \beta_4 b_m^2 + \beta_5 a_m b_m)}{\partial a_m} &= 0, \\ \frac{\partial(\beta_0 + \beta_1 a_m + \beta_2 b_m + \beta_3 a_m^2 + \beta_4 b_m^2 + \beta_5 a_m b_m)}{\partial b_m} &= 0. \end{aligned}$$

Solving this system of equations for a_m and b_m we get

$$a_m = \frac{2\beta_1\beta_4 - \beta_2\beta_5}{\beta_5^2 - 4\beta_3\beta_4} \quad \text{and} \quad b_m = \frac{2\beta_2\beta_3 - \beta_1\beta_5}{\beta_5^2 - 4\beta_3\beta_4}.$$

These can be estimated as follows.

```
dmethod(m, pfunc = "(2*b1*b4 - b2*b5)/(b5^2 - 4*b3*b4)",
  pname = c("b0", "b1", "b2", "b3", "b4", "b5"))
```

estimate	se	lower	upper	tvalue	df	pvalue
86.86	2.738	81.5	92.23	31.72	Inf	7.422e-221

```
dmethod(m, pfunc = "(2*b2*b3 - b1*b5)/(b5^2 - 4*b3*b4)",
  pname = c("b0", "b1", "b2", "b3", "b4", "b5"))
```

estimate	se	lower	upper	tvalue	df	pvalue
176.7	3.759	169.3	184	47	Inf	0